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1C - TNK

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Tugas 3

1. An electric dipole consists of charges +2e and -2e separated by 0,78 nm. It is in an electric field of strength $3,4 \times 10^6$ N/C. Calculate the magnitude of the torque on the dipole when the dipole moment is (a) parallel to, (b) perpendicular to, and (c) antiparallel to the electric field.

Jawab: (a) Parallel to: $\sin 0° = 0$

$$\vec{\tau} = \vec{p} \times \vec{E}$$

$$|\vec{\tau}| = |\vec{p}| \times |\vec{E}| \sin\theta$$

$$= (2e)(0,78\ nm)(3,4 \times 10^6\ N/C) \times (0)$$

$$= 0 //$$

(b) Perpendicular to: $\sin 90° = 1$

$$\vec{\tau} = \vec{p} \times \vec{E}$$

$$|\vec{\tau}| = |\vec{p}| \times |\vec{E}| \sin\theta$$

$$= (2e)(0,78\ nm)(3,4 \times 10^6\ N/C) \times (1)$$

$$= (2(1,6 \times 10^{-19}\ C))(0,78 \times 10^{-9}\ m)(3,4 \times 10^6\ N/C)$$

$$= 8,48 \times 10^{-22}\ Nm //$$

(c) Antiparallel to: $\sin 180° = 0$

$$\vec{\tau} = \vec{p} \times \vec{E}$$

$$|\vec{\tau}| = |\vec{p}| \times |\vec{E}| \sin\theta$$

$$= (2e)(0,78\ nm)(3,4 \times 10^6\ N/C) \times (0)$$

$$= 0 //$$

2. An electric dipole consisting of charges of magnitude 1,50 nC separated by 6,20 μm is in an electric field of strength 1100 N/C. What are (a) the magnitude of the electric dipole moment and (b) the difference between the potential energies for dipole orientations parallel and antiparallel to $\vec{E}$?

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Jawab : (a) $\vec{P} = q \times d$

$= (1,5 \times 10^{-9} C) \times (6,2 \times 10^{-6} m)$

$= 9,3 \times 10^{-15} Cm$ //

(b) $V = q \times d \times \cos\theta - q \times d \times \cos\theta$

$= q \times d \times \cos(0^\circ) - q \times d \times \cos(180^\circ)$

$= qd(1) - qd(-1)$

$= qd + qd$

$= 2qd \Rightarrow 2P$

$V = 2P$

$= 2(9,3 \times 10^{-15} Cm)$

$= 18,6 \times 10^{-15} Cm$ //

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